

Topic 4 : Aggregation

- Market Demand
- Market Supply

ECON 101
Sp 23



From individual to aggregate demand

→ Individual : understood broadly →

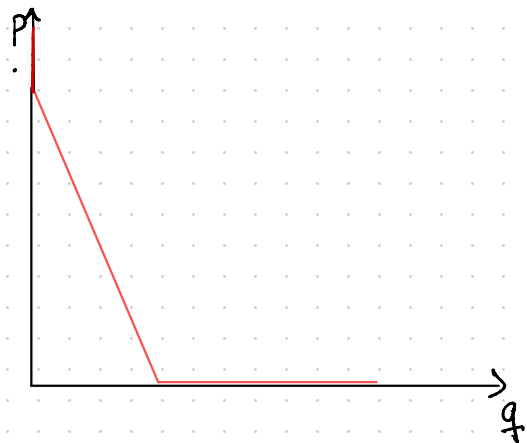
- individual consumers
- regional demand
- country demand
- $\vdots$

$q_i^D(p)$ for individual i (all other determinants are held fixed)

Remember that demand is only defined for positive quantities

↳ a convenient way to write this is

$$\max \{q_i^D(p), 0\} \Leftrightarrow q_i^D(p) \geq 0$$



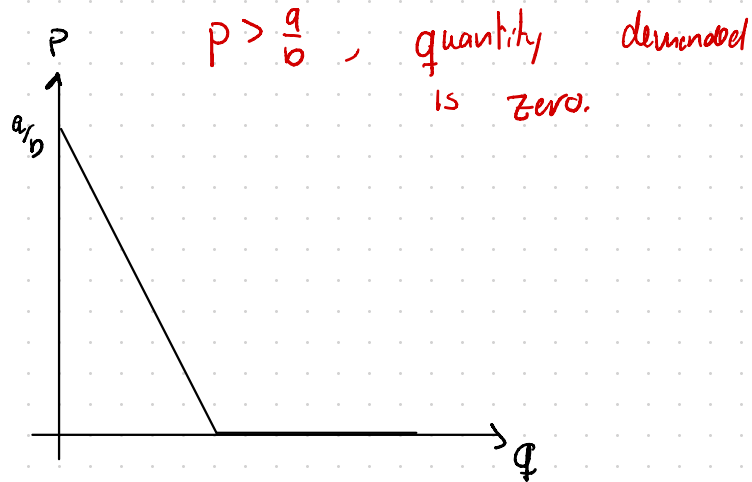
Consider a linear demand curve: $q_i^D(p) = a - bp$

At what prices is $q_i^D(p) \geq 0 \Leftrightarrow \max\{a - bp, 0\} = \begin{cases} a - bp, & \text{if } p \in [0, \frac{a}{b}] \\ 0 & \text{otherwise.} \end{cases}$

$\Leftrightarrow a - bp \geq 0 \Leftrightarrow \frac{a}{b} \geq p$, together with prices being positive leads to define the demand curve only for

$$0 \leq p \leq \frac{a}{b}$$

Inverse demand: $p = \frac{a}{b} - \frac{1}{b} q_i^D$



Market Demand (for good @)

Suppose we have the demands for $i=1, \dots, n$

$$q_1(p), q_2(p), \dots, q_n(p)$$

Market demand is defined as the **Horizontal Sum** of each individual demand: For any price I will add up all the individual demands

$$Q^D(p) = \max\{q_1(p), 0\} + \max\{q_2(p), 0\} + \dots + \max\{q_n(p), 0\}$$

Remarks: say that a price $\bar{p}$, at least one of the demands becomes negative, the $\max\{\cdot, 0\}$ takes care of this problem

Example: Identical Linear demands

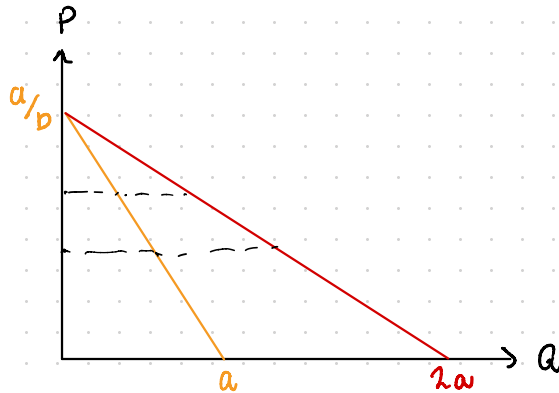
$q_i(p) = a - bp$ for $i=1, \dots, n \Rightarrow$ all demand curves $i=1, \dots, n$ are defined over $0 \leq p \leq \frac{a}{b}$

Then the market demand is: $Q(p) = n(a - bp) = na - nbp$
for $0 \leq p \leq \frac{a}{b}$

Suppose $n=2$; $Q(p) = 2a - 2bp$

$\hookrightarrow$ Inverse demand (mkt) $\Rightarrow \frac{2a - Q}{2b} = p$

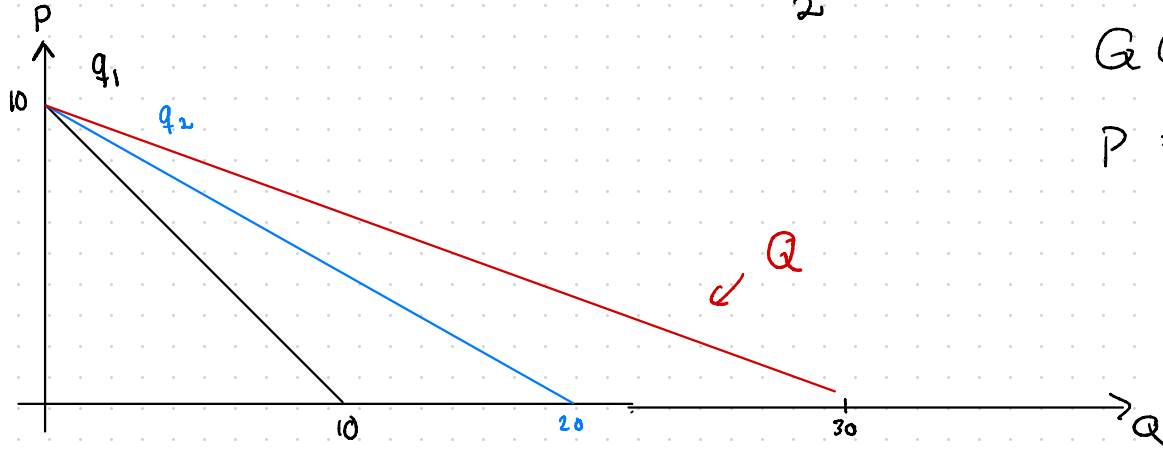
$$p = \frac{a}{b} - \frac{1}{2b} Q$$



Example:

$$q_1(p) = 10 - p,$$

$$q_2(p) = 20 - 2p$$



Inverse demands

$$p = 10 - q_1$$

$$p = 10 - \frac{q_2}{2}$$

$$q_1(p) = 10 - p,$$

$$q_2(p) = 20 - 2p$$

$$Q(p) = 30 - 3p$$

$$p = 10 - \frac{Q}{3}$$

Example:

$$\left. \begin{array}{l} q^{\text{domestic}}(p) = 1430 - 55p \\ q^{\text{foreign}}(p) = 1470 - 70p \end{array} \right\} \Rightarrow Q^{\text{world}}(p) = \max \{1430 - 55p, 0\} \\ \max \{1470 - 70p, 0\}$$

$$q^d(p) \geq 0 \Leftrightarrow 1430 - 55p \geq 0 \Leftrightarrow \frac{1430}{55} = 26 \geq p$$

$$q^f(p) \geq 0 \Leftrightarrow 1470 - 70p \geq 0 \Leftrightarrow \frac{1470}{70} = 21 \geq p$$

$$Q^w(p) = \begin{cases} 0 & p > 26 \\ 1430 - 55p & 21 < p \leq 26 \\ 2900 - 125p & 0 \leq p \leq 21 \end{cases}$$

$$\left. \begin{aligned} q^{\text{domestic}}(p) &= 1430 - 55p \\ q^{\text{foreign}}(p) &= 1470 - 70p \end{aligned} \right\} \Rightarrow Q^{\text{world}}(p) = \max \{1430 - 55p, 0\} + \max \{1470 - 70p, 0\}$$

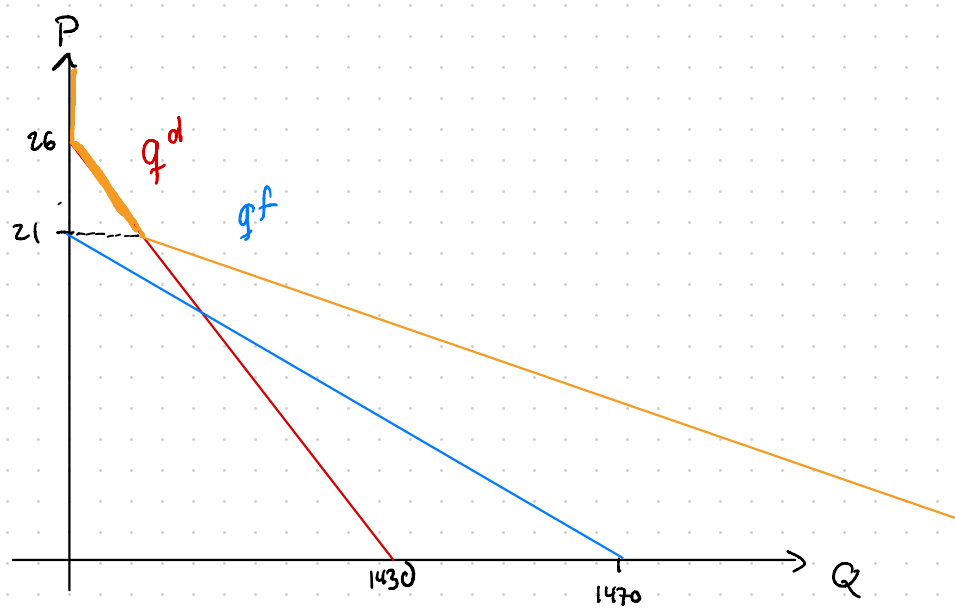
$$q^d(p) \geq 0 \Leftrightarrow 1430 - 55p \geq 0 \Leftrightarrow \frac{1430}{55} = 26 \geq p$$

$$q^f(p) \geq 0 \Leftrightarrow 1470 - 70p \geq 0 \Leftrightarrow \frac{1470}{70} = 21 \geq p$$

$$Q^w(p) = \begin{cases} 0 & p > 26 \\ 1430 - 55p & 21 < p \leq 26 \\ 2900 - 125p & 0 \leq p \leq 21 \end{cases}$$

For q^d $\Rightarrow p = 26 - \frac{1}{55} q^d$

q^f $\Rightarrow p = 21 - \frac{1}{70} q^f$



From individual to Aggregate Supply

- Individual $\rightarrow$ broadly (regions, states, firms, ...)
- We need to be careful to check at what set of prices is supply defined

$\downarrow$

$q_i^s(p)$: consider all determinants of supply are fixed.

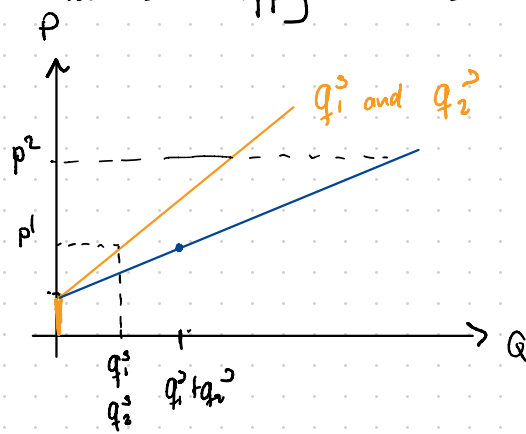
$$\hookrightarrow q_i^s(p) \geq 0 \iff \max \{ q_i^s(p), 0 \}$$

Aggregate Supply

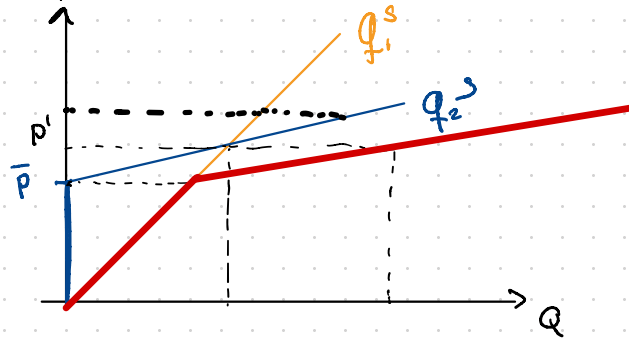
$$Q^s(p) = \max \{ q_1^s(p), 0 \} + \max \{ q_2^s(p), 0 \} + \dots + \max \{ q_n^s(p), 0 \}$$

Graphical Examples

- Identical supply curves



- Different Supply curves



- Domestic Supply / Foreign supply (Import quota)

